Make a Model of the Earth
Project Assignment Sheet
50 points

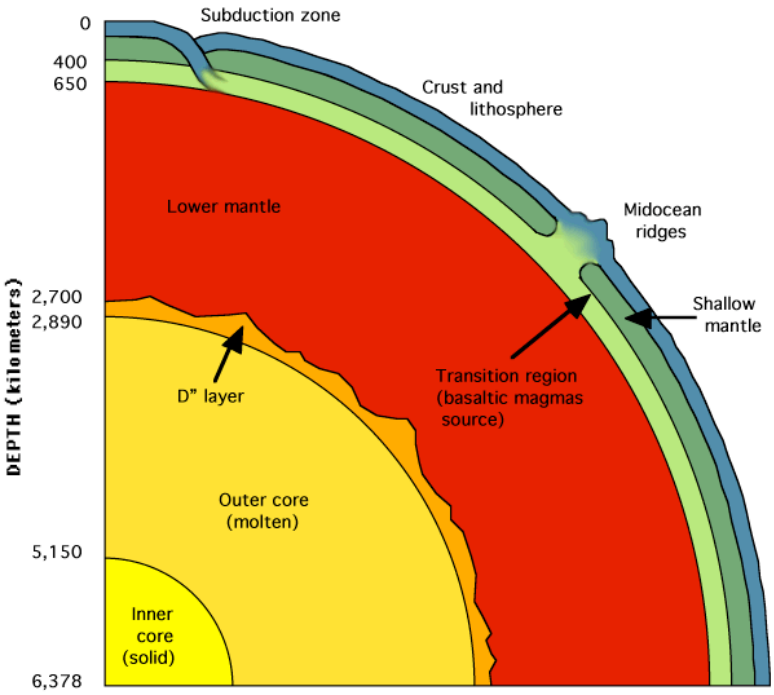
How can you make a scale model that shows both what's inside planet Earth and how the inside affects features on the surface? You will design and create such a model as a graded project. Be colorful and creative! Read the following carefully and fulfill each criteria:

Model Criteria and Grading:

- Model must have a scale and all parts must be proportional to that scale. (show the scale on an information card including your name) (5 points)
- Must include a cut-away to show (proportionally) the interior with all of the layers of Earth labeled.
  Show which layers have convection currents.
  The cut-away not to exceed 1/3 of the model. (10 points)
- Include at least three of the tectonic plates (labeled) that form Earth's surface including landmasses or continents (10 points)
- Include the ability of the plates **to move** to show how the plates push together (converge), pull apart (diverge) or slide past each other (diverge) and indicate their direction of movement (10 points with possible bonus points)
- Land features such as mountains, volcanoes, sea-floor spreading or other appropriate features created by plate movement (10 points)
- Label the parts such as layers, land features, trench or sea-floor spreading and tectonic plate (3 points)

You may use any materials that are safe and appropriate such as clay, salt dough, modeling clay, magnets, wood, Styrofoam, paper mache or other non-food materials. Make it as realistic looking as possible. However, the diameter of the earth should not exceed 2 feet due to space constraints in the classroom.

Project Due Date: _____
- No Exceptions



Future of the Earth's Surface Model

Project Assignment Sheet

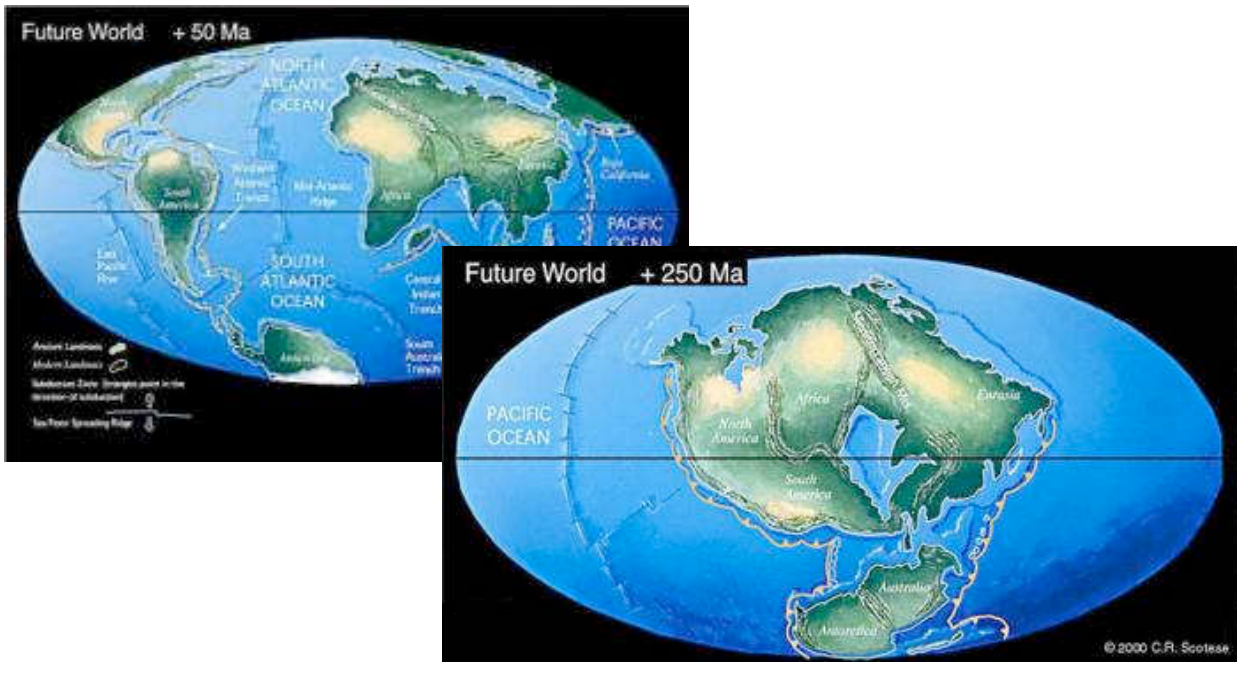
50 points

Using what you know of continental drift, convection currents, plate tectonics, and the historical movement of the continents, make a 3-dimensional Earth model (sphere or flat) showing how Earth's surface will be 100 -150 millions years in the future! Be creative, but base your model on scientific research. (not science fiction) (Pages 143 – 147 & 162 of your science book may be a good reference, also check the website and Tube videos listed below)

Criteria:

- Model must have a scale and all parts must be proportional to that scale. (show the scale on an information card including your name) (5 points)
- Outline the tectonic plates and indicated with arrows the direction of movement. (10 points)
- Include all of the continents and their new location. Clearly label. (5 points)
- Land forms or other surface features that are created due to tectonic plate movement, for example volcanoes, mountain chains, mid ocean ridges, rift valleys, deep ocean trench. Clearly label. (10 points)
- Model minimum size is 12in. x 12in., but not to exceed 18in. x 18 in.
- You may use any materials that are safe and appropriate such as clay, salt dough, modeling clay, magnets, wood, Styrofoam, paper mache or other non-food materials. Make it as realistic looking as possible.

Helpful animation site: http://www.scotese.com/futanima.htm
NASA Continents in Collision: Pangea Ultima:
http://science.nasa.gov/science-news/science-at-nasa/2000/ast06oct_1/
You Tube: Earth 100 Million Years From Now
Your Tube: 650 Million Years in under 2 minutes

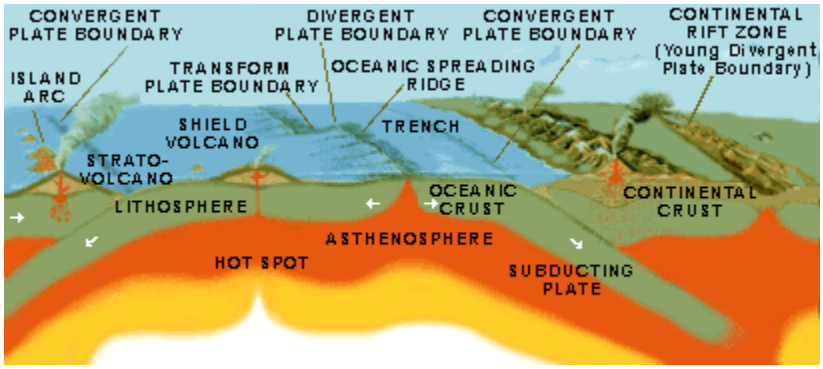


Tectonic Plate Boundaries Model
Project Assignment Sheet
50 points

Using what you have studied and know about the 3 types of tectonic plate boundaries, create a 3-dimensional model or puzzle that simulates the plate movements both on continental crust and oceanic crust. (Pgs. 149 ~ 155 of your textbook may be helpful.)

Model Criteria and Grading:

- Model has an information card including your name, and class period (5 points)
- Include the three types of tectonic plates boundaries on both continental crust and oceanic crust (labeled) (10 points)
- Must include a top view that includes the typical land forms created at that type of plate boundary, for example, volcanoes, rift valley, mid-ocean ridge, deep ocean trench (10 points)
- A side view to illustrate the movement of plates (10 points)
- Include the ability of the plates **to move** to show how the plates push together (converge), pull apart (diverge) or slide past each other (transform) and use arrows indicate their direction of movement (10 points with possible bonus points)
- Model minimum size is 12in. x 12in., but not to exceed 18in. x 18 in.
- Be colorful and creative and detailed (5 points)



You may use any materials that are safe and appropriate such as clay, salt dough, modeling clay, magnets, wood, Styrofoam, paper mache or other non-food materials. Make it as realistic looking as possible.

Project Due Date: _____
- No Exceptions